Venkateswara Rao Jannegorla

mrjvenkyofficial18@gmail.com | +1-6567891742 | Tampa, FL | portfolio | github | linkedin

Summary

Results-driven **AI/ML Engineer** with 2+ years of experience designing and deploying **large-scale ML systems**, **LLM-powered pipelines**, and **production-grade RAG and generative AI applications**. Published Python package author (**DocuWeave**, **2.9× faster** RAG ingestion vs. LangChain), IEEE/Scopus-published researcher, and open-source contributor . Deep expertise across the full ML lifecycle: data engineering, model fine-tuning (LoRA/PEFT), vector retrieval, REST API deployment, agentic AI systems, and executive-facing business intelligence. Passionate about turning cutting-edge AI research into robust, scalable production systems. Pursuing M.S. in AI & Business Analytics, University of South Florida (Dec 2026).

Technical Skills

Languages & Frameworks: Python, SQL, R, JavaScript, TypeScript; FastAPI, Flask, Next.js, Node.js, Express, React

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, Keras, HuggingFace Transformers, PEFT, LoRA, BERT, GPT, Diffusers, OpenCV

LLMs & Generative AI: LLM Fine-Tuning, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Engineering, LangChain, LlamaIndex

Data Engineering & MLOps: Pandas, NumPy, SciPy, PySpark, Apache Spark, Kafka, Hadoop, Docker

Databases & Vector Stores: PostgreSQL, MySQL, MongoDB,Pinecone, Weaviate, Qdrant, FAISS

Cloud & Infrastructure: AWS (EC2, Lambda, S3), Azure, Docker, Cloudflare, Vercel

Analytics & Statistics: A/B Testing, Hypothesis Testing, Anomaly Detection, Time-Series Analysis, Feature Engineering, Power BI, Tableau

Open Source & Research Impact

DocuWeave (PyPI Published) — Layout-aware PDF parsing and RAG-optimized chunking library; benchmarked **2.9× faster ingestion** (∼22s vs ∼65s) than LangChain at identical Recall@k (0.833). Integrates natively with LangChain, LlamaIndex, Pinecone, Weaviate, and FAISS. Preserves document semantic hierarchy and produces token-aware chunks optimized for embedding models.

Vachan AI (*In Development*) — Designing, training, and evaluating a **dialect-aware Telugu foundation LLM** with LoRA fine-tuning and distributed training on regional slang, conversational nuance, and code-mixed Telugu-English language. Covers instruction tuning, inference optimization, and evaluation across dialectal benchmarks.

Professional Experience

Vegrade Innovations — *AI/ML Engineer* Vijayawada, India | Jan 2024 – Dec 2024

- Architected and automated **end-to-end ML pipelines** ingesting and transforming **500K+ records/day** across structured and semi-structured sources; reduced data inconsistency detection latency by **40%**, directly accelerating analytical turnaround for product and business stakeholders.
- Engineered **customer churn prediction models** using behavioral, transactional, and engagement feature engineering; drove a **17% improvement in retention rate** and deployed the model as a **low-latency REST API** powering real-time production scoring for CRM teams.
- Fine-tuned **BERT-based NLP classifiers** for multi-class intent detection and document categorization; achieved a **26% F1-score gain** over baseline through systematic error analysis, class-weight balancing, learning rate scheduling, and iterative feature refinement on imbalanced corpora.
- Designed and delivered **executive-facing BI dashboards** in Tableau and Power BI, translating complex ML model outputs, drift signals, and anomaly alerts into clear, actionable business narratives for C-suite and non-technical stakeholders.
- Integrated **third-party and internal REST APIs** to continuously enrich behavioral datasets with real-time signals, enabling richer downstream feature engineering and improving generalization across diverse customer segments.
- Implemented **model monitoring and drift detection** pipelines using statistical process control techniques, reducing silent model degradation incidents and enabling proactive retraining triggers tied to business KPIs.

App Genesis — *Junior Machine Learning Engineer Intern* Vijayawada, India | Aug 2022 – Dec 2023

- Built and productionized **NLP models** for multi-label text classification and fine-grained sentiment analysis (Python, Scikit-learn, HuggingFace Transformers); exposed models via **FastAPI** with request batching, input validation, and response caching for low-latency production inference.
- Integrated **LLM-based content understanding automation** into analytics and publishing workflows, improving content relevance scoring metrics and driving a measurable **22% uplift in user engagement** across customer-facing features.
- Developed **real-time ML inference pipelines** with async request handling, reducing **prediction latency by 38%** through INT8 model quantization, dynamic batching, and architectural pruning without significant accuracy loss.
- Conducted systematic **failure case and confusion matrix analysis** across structured and unstructured datasets, identifying high-error subcategories and establishing automated feedback loops that drove iterative model improvements across multiple deployment cycles.
- Collaborated with product and engineering teams to translate business requirements into ML-driven features, owning the full lifecycle from data exploration and prototyping to model deployment and post-launch monitoring.

Projects

Financial Report RAG System Python · FAISS · spaCy · Sentence-Transformers · LLMs

- Built an **enterprise-grade RAG pipeline** for analytical Q&A over large financial report corpora using **hybrid retrieval** combining dense vector search (FAISS + Sentence-Transformers) and sparse BM25 for high-recall candidate generation, followed by LLM-based answer synthesis.
- Designed multi-stage document processing with layout-aware chunking, entity-aware metadata filtering (company, fiscal period, metric type), and cross-encoder re-ranking to significantly improve retrieval precision over naive vector search baselines.
- Implemented **faithfulness and citation grounding** checks using NLI-based verification to reduce hallucination in generated financial answers, achieving over **90% answer-source alignment** in evaluation.
- Engineered a **query understanding layer** that classifies incoming questions by financial intent (trend, comparison, ratio, risk) and routes them to specialized retrieval strategies, improving end-to-end answer quality on complex multi-step financial reasoning tasks.

AI Data Analysis Agent LLaMA · Gemini AI · Python · Pandas · LLMs · Automation

- Designed and deployed an **end-to-end agentic data analysis system** powered by LLaMA and Gemini AI that autonomously ingests CSV datasets, performs **automated EDA** (distributions, correlations, outliers, missing values), and generates dynamic visualizations with minimal user input.
- Built a **multi-step reasoning loop** with tool-use capabilities: the agent selects analysis strategies, writes and executes Python code, interprets outputs, and iteratively refines insights before composing a **comprehensive narrative report** with charts, statistical summaries, and recommendations.
- Implemented **structured output schemas** and prompt chaining to ensure consistent, auditable agent behavior across diverse dataset types including sales, operations, healthcare, and financial data.
- Deployed as a **full-stack web application** (Next.js + FastAPI backend) with streaming agent outputs, interactive chart rendering, and one-click PDF report export; live at ai-data-analysis-agent-zeta.vercel.app.

Ops Copilot — AI Operations Assistant Python · LLMs · Agentic AI · Automation · FastAPI

- Engineered an **AI-powered operations copilot** for intelligent task automation and anomaly triage in DevOps and SRE workflows; the system ingests real-time metrics, logs, and alert streams, then applies **agentic reasoning loops** to autonomously diagnose infrastructure anomalies and surface root-cause hypotheses with supporting evidence.
- Built a **conversational runbook execution interface** allowing on-call engineers to query system state, trigger remediation actions, and receive plain-language incident summaries, significantly reducing mean time to resolution (MTTR) in simulated incident scenarios.
- Integrated **tool-use and memory** capabilities into the agentic loop, enabling the copilot to retain incident context across multi-turn interactions, correlate patterns across historical events, and proactively suggest preventive actions.
- Designed a **pluggable integration layer** supporting ingestion from Prometheus, Grafana, PagerDuty, and custom log streams via a unified FastAPI adapter, making the system deployable across heterogeneous observability stacks without code changes.

VARAG — Vision-Augmented RAG LLaVA · LlamaIndex · Qdrant

- Developed a **multimodal RAG system** combining LLaVA-based visual feature extraction with text embeddings in a unified retrieval pipeline, enabling joint reasoning over mixed document types (PDFs with figures, slides, scanned images, and tables); improved contextual precision by **60%** over text-only baselines.
- Engineered a **modality-aware indexing strategy** in Qdrant using separate vector spaces for text and image embeddings with late-fusion reranking, allowing the system to dynamically weight visual vs. textual evidence based on query semantics.
- Built an **image captioning and OCR pre-processing pipeline** to extract structured text from figures, charts, and scanned pages before embedding, significantly improving retrieval recall on visually-dense documents such as scientific papers and technical manuals.
- Evaluated system performance using a **custom multimodal QA benchmark** constructed from mixed-modality documents, measuring Recall@k, MRR, and answer faithfulness independently for text-only, image-only, and fused retrieval conditions.

Publications

Phishing Detection: A Predictive Model for Cyber Security — *IEEE, 2023.* Developed an ensemble ML model leveraging URL structure, page content features, and WHOIS metadata for high-precision phishing site identification; demonstrated superior accuracy over prior rule-based approaches on benchmark cybersecurity datasets.

Secure Document Storage System Using Web3 — *Scopus Indexed, 2024.* Designed a blockchain-based decentralized document storage architecture using smart contracts and distributed ledger technology to ensure tamper-proof integrity, access-controlled sharing, and cryptographically verifiable audit trails for sensitive organizational documents.

Education

University of South Florida — *M.S. Artificial Intelligence & Business Analytics* Tampa, FL | Expected Dec 2026

Lakireddy Balireddy College of Engineering — *B.Tech. Artificial Intelligence & Data Science* India |